SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:
Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL6)

Assessment Tool Weightage: 30%

19/20

QUESTIONS: 5 **Total Marks:25**

Annexure A

Comparative Analysis

Code of Conduct: 192472056

I V.sheerthika / 192472056 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Comparative Analysis

1. Star Topology vs Bus Topology – Cost, Reliability and Performance

Introduction

Star and bus are two important network topologies. Both can be used to connect multiple devices, but their architecture and operational characteristics are significantly different.

In a star topology, every computer or network device has a dedicated connection to a central switch or hub. The central device controls communication between the connected nodes.

In a bus topology, all devices share a common backbone communication cable. Data transmitted by one device travels along the backbone, and the appropriate destination device receives the data.

Cost Comparison

Bus topology is generally cheaper than star topology because it requires less cabling and fewer networking devices. A single backbone cable can connect multiple devices.

Star topology requires a separate cable connection between every device and the central switch. It also requires a switch or hub, increasing the initial investment.

However, the initial cost is not the only factor that should be considered. Star topology may have a lower total maintenance cost because faults can be identified and repaired more easily.

Reliability

Star topology provides better reliability than bus topology. If the cable connecting one workstation to the switch fails, only that particular workstation loses connectivity. Other devices continue to communicate normally.

In bus topology, the backbone cable is shared by all devices. If the backbone is damaged, the entire network can be affected.

Therefore, star topology is more suitable for organizations where network availability is important.

Performance

Star topology generally provides better performance because modern switches can provide dedicated communication paths between devices. This reduces unnecessary traffic and collisions.

Bus topology uses a shared communication medium. When several devices attempt to transmit data simultaneously, collisions can occur. As the number of connected devices increases, network traffic also increases and performance may decrease.

Troubleshooting

Troubleshooting is easier in a star network because each device has an individual connection. A network administrator can check a specific switch port or cable to identify the problem.

In a bus network, identifying a fault in the backbone can be difficult because a single cable problem can affect many devices.

Example

Consider a company with 20 employees. Each employee has a computer, and the company also has printers, IP phones and access points. A star topology with a managed switch is a suitable choice because individual failures can be isolated and the network can be expanded easily.

Conclusion

Although bus topology is cheaper and simpler, star topology provides better performance, reliability, scalability and maintenance. Therefore, star topology is generally more suitable for modern office and enterprise LANs, while bus topology is mainly useful for understanding traditional networking concepts and certain specialized or legacy environments.

2. Mesh Topology vs Star Topology – Fault Tolerance

Introduction

Fault tolerance is the ability of a network to continue operating even when one or more components fail. This is particularly important for organizations such as banks, hospitals, data centers and financial institutions.

Mesh and star topologies provide different levels of fault tolerance.

Mesh Topology

In a mesh topology, devices are connected through multiple links. In a full mesh, every device has a direct connection to every other device.

For n devices, the number of links required in a full mesh is:

$$L = \frac{n(n-1)}{2}$$

For example, if there are 5 devices:

$$L = \frac{5(5-1)}{2} = 10$$

Therefore, a full mesh requires a large number of connections as the network grows.

Fault Tolerance of Mesh.

The major advantage of mesh topology is redundancy. If one communication link fails, another path can be used to transmit data.

For example, suppose Router A normally communicates with Router B through one link. If that link fails, traffic can potentially travel through Router C and reach Router B.

This makes mesh topology highly suitable for critical infrastructure.

Fault Tolerance of Star

Star topology also provides a degree of fault tolerance. If one workstation or cable fails, the other devices continue operating.

However, the central switch is an important point of failure. If the central switch stops working and there is no redundancy, communication between connected devices can be interrupted.

Cost Comparison

Mesh topology requires significantly more cables, ports and networking equipment. Therefore, installation and maintenance costs are much higher.

Star topology is more economical because each device only needs one connection to the central switch.

Applications

Mesh topology or partial mesh designs are commonly used for:

- Banking networks
- Data center core networks
- Internet service provider networks
- Critical enterprise networks
- Disaster recovery infrastructure

Star topology is commonly used for:

- Offices
- Computer laboratories
- Schools
- Small businesses
- Departmental LANs

Conclusion

Mesh topology provides superior fault tolerance because of its multiple communication paths. Star topology provides good reliability at a much lower cost and with simpler management.

Therefore, mesh topology is preferred for mission-critical networks, whereas star topology is more practical for ordinary LAN environments.

3. Bus Topology vs Mesh Topology – Installation Complexity

Introduction

Installation complexity is an important consideration when designing a network. A topology that is easy to install reduces deployment time, labor requirements and initial cost.

Bus and mesh topologies represent two very different approaches to network installation.

Bus Topology

Bus topology uses one common backbone cable. Each network device connects to the backbone.

Traditional bus networks used coaxial cables, connectors and terminators. Because devices shared a common communication medium, the physical structure was relatively simple.

The main advantages are:

- Less cabling
- Low hardware requirements
- Simple initial installation
- Low cost

However, the shared backbone becomes a major limitation.

Mesh Topology

Mesh topology requires multiple connections between devices. In a full mesh, every device must have a direct link to every other device.

As the number of devices increases, the number of required links increases rapidly.

For example:

Number of Devices	Full Mesh Links
3	3
4	6
5	10
10	45
20	190

This demonstrates why a full mesh becomes difficult and expensive to install as the network grows.

Maintenance Complexity

Bus topology has relatively simple physical construction, but troubleshooting can become difficult if the backbone fails.

Mesh topology provides redundancy but has many physical and logical connections. Network administrators must monitor a much larger number of links.

Cost

Bus topology has a lower installation cost because it uses fewer cables and components.

Mesh topology has a much higher cost because every connection requires networking ports, cables and configuration.

Reliability

Although bus topology is easy to install, it has lower fault tolerance. A backbone failure can affect many or all devices.

Mesh topology is more difficult to install but provides excellent redundancy.

Conclusion

Bus topology is simpler, cheaper and easier to install, whereas mesh topology is complex and expensive but provides much higher redundancy and fault tolerance. The choice depends on the organization's priorities.

4. Hybrid Topology vs Star Topology – Scalability

Introduction

Scalability refers to the ability of a network to grow without requiring a complete redesign. This is especially important for universities, multinational companies, hospitals and large organizations.

Hybrid topology and star topology both support expansion, but hybrid topology provides greater flexibility.

Star Topology

In a star topology, all devices connect to a central switch.

For example:

```
        Central Switch
        /        |   \
      PC1      PC2   PC3
```

If additional devices are required, unused switch ports can be used.

However, once all ports are occupied, another switch must be introduced. The network therefore depends on the capacity and architecture of the central devices.

Hybrid Topology

A hybrid topology combines two or more topologies. For example, a large university can use:

- Mesh at the campus core
- Star inside buildings
- Fiber between buildings
- Cat6 connections inside floors

A simplified structure is:

```
          Campus Core
           /   |   \
          /    |    \
Building A Building B Building C
   |         |        |
 Switch    Switch    Switch
  /|\       /|\       /|\
PC PC AP  PC PC AP  PC PC AP
```

This approach allows each part of the organization to use a topology suitable for its specific requirements.

Conclusion

Hybrid topology provides greater scalability and flexibility than a simple star topology, especially for large organizations. However, star topology remains easier and cheaper to implement for small and medium-sized networks.

5. Maintenance of Star, Mesh, Bus and Hybrid Topologies

Introduction

Network maintenance includes monitoring, troubleshooting, repairing cables and devices, replacing failed components, managing configurations and ensuring network availability.

Different topologies have different maintenance requirements.

Star Topology Maintenance

Star topology is relatively easy to maintain.

Every device has a dedicated connection to the central switch. If one computer stops working, the administrator can check:

1. The computer's network interface
2. The Ethernet cable
3. The switch port
4. The IP configuration

A failure normally affects only one device.

Therefore, fault isolation is one of the biggest advantages of star topology.

Mesh Topology Maintenance

Mesh topology is highly reliable but more difficult to maintain.

Because there are many connections, administrators need to monitor a larger number of links and devices.

If a link fails, redundancy can keep the network operating, but the administrator must still identify and repair the failed connection.

Mesh networks therefore require skilled network engineers and advanced monitoring tools.

Bus Topology Maintenance

Bus topology can be difficult to troubleshoot.

Since many devices share the same backbone, a problem in the main cable can affect the entire network.

The administrator must identify the location of the fault along the backbone.

This makes bus topology less attractive for large modern networks.

Hybrid Topology Maintenance

Hybrid topology has moderate to high maintenance complexity.

Different sections may use different technologies. For example:

- Fiber backbone
- Ethernet star networks
- Wireless networks
- Redundant core links
- VLANs

Administrators must understand all these technologies.

However, the flexibility of hybrid topology makes it very useful for large organizations.

Comparison

Factor	Star	Mesh	Bus	Hybrid
Fault isolation	Excellent	Moderate/Complex	Poor	Good
Maintenance	Easy	Difficult	Difficult	Moderate
Troubleshooting	Easy	Complex	Difficult	Moderate
Reliability	High	Very High	Low	High
Cost	Moderate	Very High	Low	Variable
Scalability	High	Limited by cost/complexity	Low	Very High

Conclusion

Star topology provides the best balance between simplicity, reliability and maintenance for most LANs. Mesh topology provides the highest fault tolerance but requires greater technical expertise and cost. Bus topology is inexpensive but difficult to troubleshoot when failures occur. Hybrid topology provides flexibility and scalability but requires skilled network administration.

RUBRICS FOR CALCULATION

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding of Concepts	15%	Demonstrates complete and in-depth understanding of all concepts being compared	Good understanding with minor gaps	Basic understanding; some concepts unclear	Poor understanding of concepts
Comparison Depth	20%	Provides detailed, insightful, and meaningful comparisons across multiple aspects	Adequate comparison with relevant points	Limited comparison; lacks depth	Minimal or incorrect comparison
Criteria Selection	10%	Selects highly relevant and appropriate comparison parameters	Mostly relevant criteria	Some irrelevant or missing criteria	Poor or no criteria selection
Analytical Skills	15%	Strong analysis with logical reasoning and clear justification	Good analysis with some justification	Basic analysis with weak reasoning	No clear analysis
Organization & Structure	10%	Well-organized, clear, and logically structured	Generally organized	Somewhat disorganized	Poorly structured
Use of Examples / Evidence	10%	Uses strong, relevant examples or data to support comparison	Uses some examples	Limited examples	No supporting examples
Clarity of Presentation	10%	Very clear, concise, and	Mostly clear	Somewhat unclear	Difficult to understand